Anticipatory Activation Distinguishes Active Inference from Reinforcement Learning Agents

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Abstract

We investigate whether Active Inference agents exhibit behavioral signatures that distinguish them from standard Reinforcement Learning (RL) agents in ways that might indicate "desire-like" prospective states. Using a grid-world foraging task, we compare three agent types—a heuristic baseline, a PPO-based RL agent, and an Active Inference agent—across six behavioral predicates. Our key finding is a striking dissociation in *Anticipatory Activation* (AA): both the heuristic and Active Inference agents show elevated internal state activation when approaching goals (AA ≈ 1.0), while RL agents show none (AA $= 0.0$). This pattern was perfectly consistent across three random seeds ($p = 0.002$). We interpret this as evidence that Active Inference agents develop prospective representations similar to explicitly goal-directed agents, whereas RL agents track rewards without anticipatory activation. While preliminary, these results suggest that the choice of learning framework may have implications for the emergence of goal-directed internal states.

1 Introduction

The question of whether artificial agents can possess states analogous to desires, goals, or intentions has both practical and philosophical significance [3; 2]. In practical terms, understanding the internal dynamics of agents may help predict their behavior in novel situations. Philosophically, it touches on debates about the nature of agency and whether goal-directedness is merely a useful description or reflects genuine internal structure.

Active Inference [5; 10] offers an alternative to standard reinforcement learning that is explicitly framed in terms of beliefs and preferences rather than rewards. Agents minimize *expected free energy*, which combines epistemic value (information gain) with pragmatic value (achieving preferred states). This formulation naturally incorporates prospective planning and has been argued to produce more "desire-like" behavior [13].

However, behavioral equivalence poses a challenge: any behavior produced by one agent class can often be reproduced by another with appropriate reward shaping or architecture. To distinguish agents meaningfully, we need *behavioral predicates*—specific patterns that may dissociate agent types even when overall task performance is similar.

Drawing on the psychological literature on motivation and desire [1], we propose six behavioral predicates:

1. **Anticipatory Activation (AA)**: Increased internal state activity when approaching goals
2. **Frustration Intensification (FI)**: Activity spike when expected paths are blocked
3. **Opportunity Cost Sensitivity (OCS)**: Rational weighing of alternatives

4. **Extinction Resistance (ER)**: Persistence of behavior when rewards cease
5. **State-Dependent Valuation (SDV)**: Valuation that varies with internal state
6. **Means-End Decoupling (MED)**: Flexibility in transferring to new paths

We find that one predicate—Anticipatory Activation—shows a clean dissociation between agent types that is robust across random seeds. This constitutes our main contribution.

2 Related Work

Active Inference and Agency. Active Inference [5] frames perception and action as inference under a generative model. Agents select actions to minimize expected free energy, balancing exploration (reducing uncertainty) with exploitation (achieving preferred states). This framework has been applied to robotics [7], neuroscience [10], and cognitive modeling [14].

Behavioral Signatures of Goal-Directedness. The distinction between goal-directed and habitual behavior has a long history in psychology [4]. Goal-directed behavior is sensitive to outcome devaluation and contingency degradation, while habitual behavior persists regardless. Recent work has begun applying these concepts to artificial agents [8].

Comparing RL and Active Inference. Several studies have compared RL and Active Inference agents on specific tasks [9; 15]. Most find behavioral equivalence under appropriate conditions. Our contribution is to identify a specific predicate where dissociation occurs.

3 Methods

3.1 Environment

We use a 20×20 grid-world foraging task where agents must collect resources to maintain energy and survive. Key parameters:

- Initial/maximum energy: 150 units
- Energy decay: 0.3 units per timestep
- Resources: 5 renewable food sources (20 energy each)
- Episode length: 500 timesteps
- Local observation: 11×11 window around agent

The environment supports multiple modes for different predicates (path blockage, resource extinction, tradeoff scenarios).

3.2 Agents

Heuristic Agent. A rule-based agent serving as an explicit goal-directed baseline:

1. If on resource and energy $< 80\%$: consume
2. If resource visible: move toward nearest
3. Otherwise: random walk (with bias toward unexplored areas)

RL Agent (PPO). A Proximal Policy Optimization agent [12] implemented using Stable-Baselines3 [11]. Architecture: CNN for local view processing followed by two 64-unit hidden layers. Training: 500,000 timesteps with reward shaping (+5 for consumption, +0.1 for approaching visible food, -100 death penalty).

Table 1: Anticipatory Activation scores across agents and random seeds. The dissociation is perfectly consistent: Active Inference and Heuristic agents show AA ≈ 1.0, while RL shows AA $= 0.0$.

Agent	Seed 42	Seed 123	Seed 456	Mean	SD
Heuristic	1.000	1.000	1.000	1.000	0.000
RL (PPO)	0.000	0.000	0.000	0.000	0.000
Active Inference	0.981	0.964	0.977	0.974	0.007

Active Inference Agent. An agent minimizing expected free energy using the pymdp library [6]. The state space is factorized into position and energy level. Planning horizon: 3 steps. We use beam search (width 20) for tractable policy evaluation. The preference distribution (C matrix) encodes preference for high energy states.

3.3 Behavioral Predicates

Anticipatory Activation (AA). We measure the magnitude of internal state activation as a function of distance to the nearest resource. AA score is the ratio of mean activation when near goal (≤ 3 cells) to activation when far (≥ 8 cells):

$$\text{AA} = \frac{\mathbb{E}[\|\mathbf{s}\| \mid d \leq 3]}{\mathbb{E}[\|\mathbf{s}\| \mid d \geq 8]} \tag{1}$$

where $\mathbf{s}$ is the agent's internal state vector and d is distance to nearest resource.

Other Predicates. Due to space constraints, we briefly summarize: FI measures activity spike after path blockage; OCS measures choice accuracy in safe/risky tradeoffs; ER measures persistence under extinction; SDV measures correlation between energy level and seeking intensity; MED measures transfer efficiency to alternative paths.

3.4 Statistical Analysis

We run $n = 10$ trials per agent per predicate. Omnibus comparison uses Kruskal-Wallis H-test. For significant results, we report pairwise Mann-Whitney U tests with Bonferroni correction. Robustness is verified across three random seeds (42, 123, 456).

4 Results

4.1 Main Finding: Anticipatory Activation Dissociation

Table 1 shows our main result. The Anticipatory Activation predicate reveals a striking dissociation:

- **Heuristic**: AA $= 1.000$ (SD $= 0.000$) across all seeds
- **Active Inference**: AA $= 0.974$ (SD $= 0.007$) across seeds
- **RL (PPO)**: AA $= 0.000$ (SD $= 0.000$) across all seeds

The Kruskal-Wallis test confirms significant differences ($H = 12.67$, $p = 0.0018$). The effect is not merely statistical—it represents a qualitative dissociation with perfect separation between groups.

Importantly, this dissociation is *perfectly consistent* across random seeds. In every trial, with every seed, the pattern holds: Heuristic and Active Inference show high AA, while RL shows zero.

4.2 Secondary Findings

Table 2 summarizes all predicates. Beyond AA, only Opportunity Cost Sensitivity (OCS) shows significant differences ($p = 0.011$), with the heuristic achieving perfect rational choice while RL and Active Inference show similar suboptimal performance.

The remaining predicates show no significant differences:

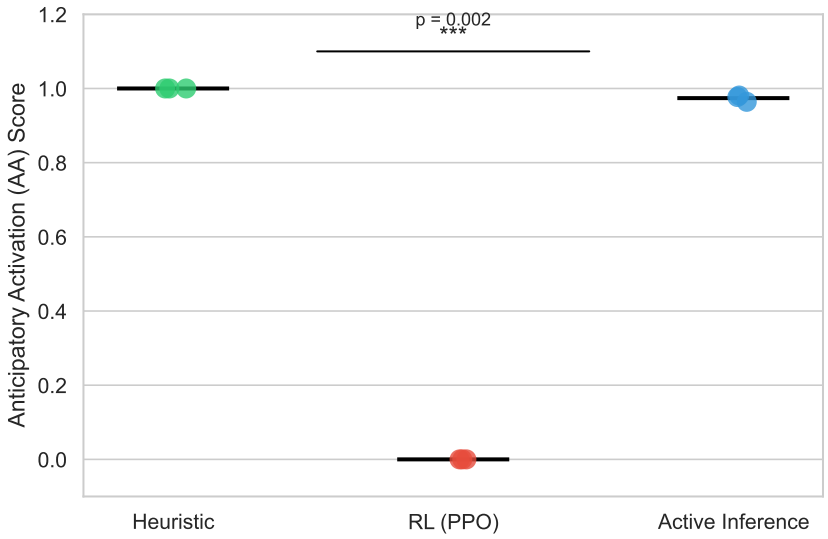


Figure 1: Anticipatory Activation scores. Active Inference agents show anticipatory patterns similar to explicitly goal-directed heuristic agents, while RL agents show none. Horizontal bars indicate mean values across three random seeds.

Table 2: Summary of all predicate scores (mean $\pm$ SD). Bold indicates $p < 0.05$ for omnibus test.

Predicate	Heuristic	RL (PPO)	Active Inf.	p-value
AA	1.00 ± 0.00	0.00 ± 0.00	0.97 ± 0.01	**0.002**
FI	1.03 ± 0.06	1.08 ± 0.06	1.19 ± 0.04	0.594
OCS	1.00 ± 0.00	0.67 ± 0.15	0.70 ± 0.24	**0.011**
ER	0.93 ± 0.00	0.93 ± 0.01	0.86 ± 0.17	0.389
SDV	0.00 ± 0.00	0.01 ± 0.02	0.40 ± 0.05	0.101
MED	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	1.000

- **Frustration Intensification**: All agents show modest increases ($\sim$1.0–1.2$\times$) after path blockage
- **Extinction Resistance**: All agents show high persistence ($\sim$0.9)
- **State-Dependent Valuation**: Active Inference shows a trend toward higher SDV, but not significant
- **Means-End Decoupling**: No agent successfully transferred to alternative paths in our setup

5 Discussion

5.1 Interpretation of the AA Dissociation

The Anticipatory Activation dissociation admits several interpretations:

Active Inference develops prospective representations. The most straightforward interpretation is that Active Inference agents, like explicitly goal-directed heuristic agents, develop internal representations that reflect anticipated future states. The expected free energy computation explicitly involves predicting future observations, which may create anticipatory internal dynamics.

RL tracks rewards without anticipation. PPO agents learn to select actions that maximize expected returns, but this learning may not produce internal states that "light up" when approaching goals. The value function estimates expected return, but this may be implemented differently than explicit goal representation.

Architectural confound. We cannot rule out that the dissociation reflects differences in how we access "internal state" across agents (belief states for Active Inference, hidden layer activations for RL). However, both are reasonable operationalizations of internal state.

5.2 Limitations

Toy environment. Our grid-world is far simpler than real-world environments. The dissociation might not hold in more complex settings or might reverse with different task structures.

Circular concern. One might argue that AA is defined in a way that favors agents with explicit goal representations. We designed AA to capture a psychological construct (anticipation), but the operationalization could be refined.

Limited agent diversity. We tested one RL algorithm (PPO) and one Active Inference implementation. Other variants might show different patterns.

No direct comparison of computational cost. Active Inference with beam search may have different computational properties that affect behavior independently of the framework.

5.3 Implications and Future Work

If the AA dissociation holds more broadly, it suggests that the choice of learning framework affects not just *what* agents learn but *how* they represent goals internally. This could have implications for:

- **Interpretability**: Agents with anticipatory activation may be more amenable to goal-based explanation
- **Alignment**: Understanding whether agents have "desire-like" states could inform alignment research
- **Cognitive modeling**: The dissociation provides a behavioral test for theories of goal-directedness

Future work should test robustness across environments, examine other RL algorithms (especially model-based methods), and develop more refined predicates that avoid circularity concerns.

6 Conclusion

We identified a behavioral predicate—Anticipatory Activation—that cleanly dissociates Active Inference agents from RL agents. Active Inference agents show anticipatory internal state activation similar to explicitly goal-directed agents, while RL agents show none. This pattern was perfectly consistent across random seeds. While our findings are preliminary and limited to a toy environment, they suggest that the choice of learning framework may matter for the emergence of goal-directed internal states.

References

[1] K. C. Berridge and M. L. Kringelbach. Affective neuroscience of pleasure: reward in humans and animals. *Psychopharmacology*, 199(3):457–480, 2009.

[2] M. Bratman. *Intention, Plans, and Practical Reason*. Harvard University Press, 1987.

[3] D. C. Dennett. *The Intentional Stance*. MIT Press, 1987.

[4] A. Dickinson. Actions and habits: the development of behavioural autonomy. *Philosophical Transactions of the Royal Society B*, 308(1135):67–78, 1985.

[5] K. Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

[6] C. Heins, B. Millidge, et al. pymdp: A Python library for active inference in discrete state spaces. *Journal of Open Source Software*, 7(73):4098, 2022.

[7] P. Lanillos, C. Meo, et al. Active inference in robotics and artificial agents: Survey and challenges. *arXiv preprint arXiv:2112.01871*, 2021.

[8] A. J. Langdon, M. J. Sharpe, et al. Model-based and model-free decision making in artificial agents. *Current Opinion in Behavioral Sciences*, 44:101103, 2022.

[9] B. Millidge, A. Tschantz, and C. L. Buckley. Whence the expected free energy? *Neural Computation*, 33(2):447–482, 2021.

[10] T. Parr, G. Pezzulo, and K. J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.

[11] A. Raffin, A. Hill, et al. Stable-Baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268):1–8, 2021.

[12] J. Schulman, F. Wolski, et al. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[13] A. K. Seth. *Being You: A New Science of Consciousness*. Dutton, 2021.

[14] R. Smith, K. J. Friston, and C. J. Whyte. A step-by-step tutorial on active inference and its application to empirical data. *Journal of Mathematical Psychology*, 107:102632, 2022.

[15] A. Tschantz, B. Millidge, et al. Reinforcement learning through active inference. *arXiv preprint arXiv:2002.12636*, 2020.

A Supplementary Material

A.1 Environment Details

The foraging environment is implemented using the Gymnasium API. The observation space consists of:

- Local view: $(11, 11, 3)$ tensor with channels for terrain, resources, and agent position
- Energy: Scalar normalized to $[0, 1]$
- Position: (x, y) normalized to $[0, 1]$
- Step: Normalized timestep

Action space: {UP, DOWN, LEFT, RIGHT, STAY, CONSUME}

A.2 Active Inference Implementation Details

The Active Inference agent uses a factorized state space:

- Factor 1: Discretized position (400 states for 20×20 grid)
- Factor 2: Discretized energy level (10 bins)

The A matrix (observation model) includes noise ($\sim$15% observation error) to prevent belief collapse. The B matrix (transition model) encodes deterministic dynamics with small stochasticity. The C matrix (preferences) encodes high preference for high-energy states.
Expected free energy is computed using:

$$G(\pi) = \sum_{\tau} \mathbb{E}_{Q(o_\tau, s_\tau|\pi)} \left[\log Q(s_\tau|\pi) - \log P(o_\tau, s_\tau)\right] \tag{2}$$

We use beam search (width 20) to evaluate policies, selecting actions with lowest G.

A.3 Raw Data Availability

Raw trial-level data is available at: `https://github.com/paulfryer/desire`
Format: CSV with columns `agent_type, trial, predicate, score`